Image Enhancement

Topic 3

Image Enhancement

- Image enhancement refers to improving or sharpening the image features: edges, boundaries, contrast, and brightness.
- Objective of enhancement techniques is to transform an image into a form more suitable than the original image for a specific application.
- These methods are application-specific.
- Thus, for example, a method that is quite useful for enhancing *x*-ray images may not necessarily be the best approach for enhancing pictures of Mars transmitted by a space probe.
- Image enhancement approaches fall into two broad categories: *spatial domain* methods and *frequency domain* methods.

Image Enhancement (cont')

- The spatial domain refers to the image plane itself, and approaches in this category are based on direct manipulation of pixels in an image.
- Frequency domain processing techniques are based on modifying the Fourier Transform of an image.
- Enhancement techniques based on various combinations of methods from these two categories are not unusual.
- We concentrate more on spatial domain image enhancement in this subject.

Spatial Domain Image Enhancement

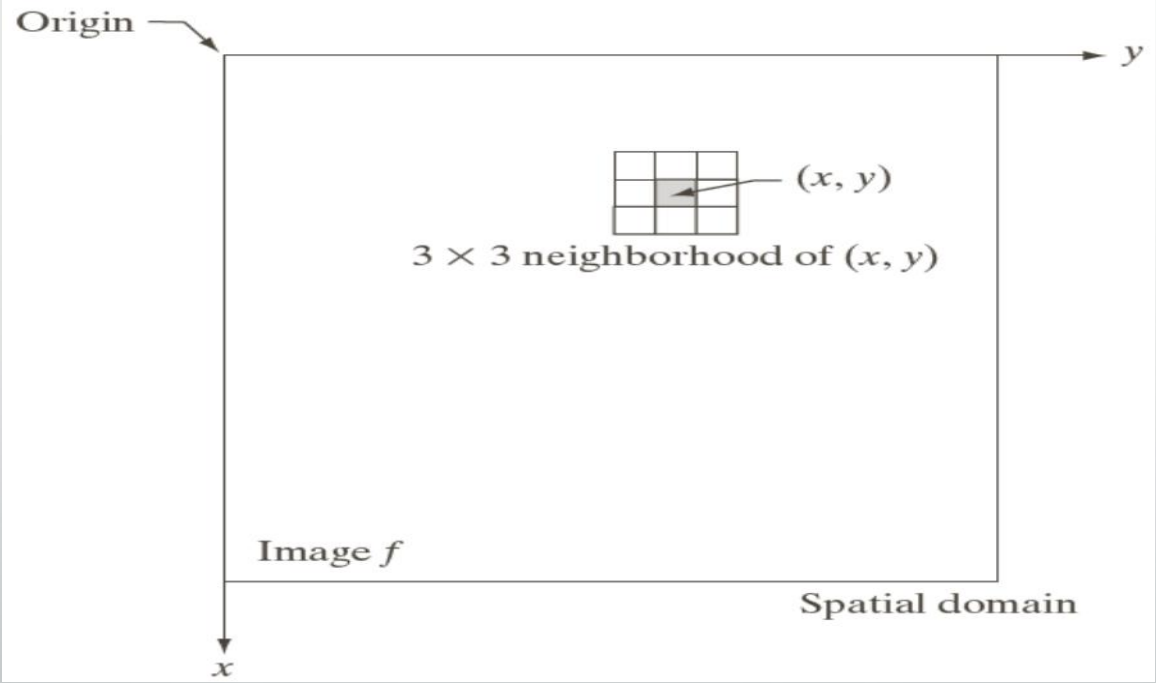
- The term spatial domain refers to the aggregate of pixels composing an image
- Spatial domain methods are procedures that operate directly on these pixels.
- It deals with the manipulation of pixel values.
- Methods of manipulation of pixel values are broadly classified into
  - Point Operation
  - Mask Operation
  - Global Operation
- Image processing, the point operation is represented as

$$g(x, y) = T[f(x, y)]$$

- Where $f(x, y)$ is the input image, $g(x, y)$ is the processed image, and T is an operator on f, defined over some neighborhood of (x, y).

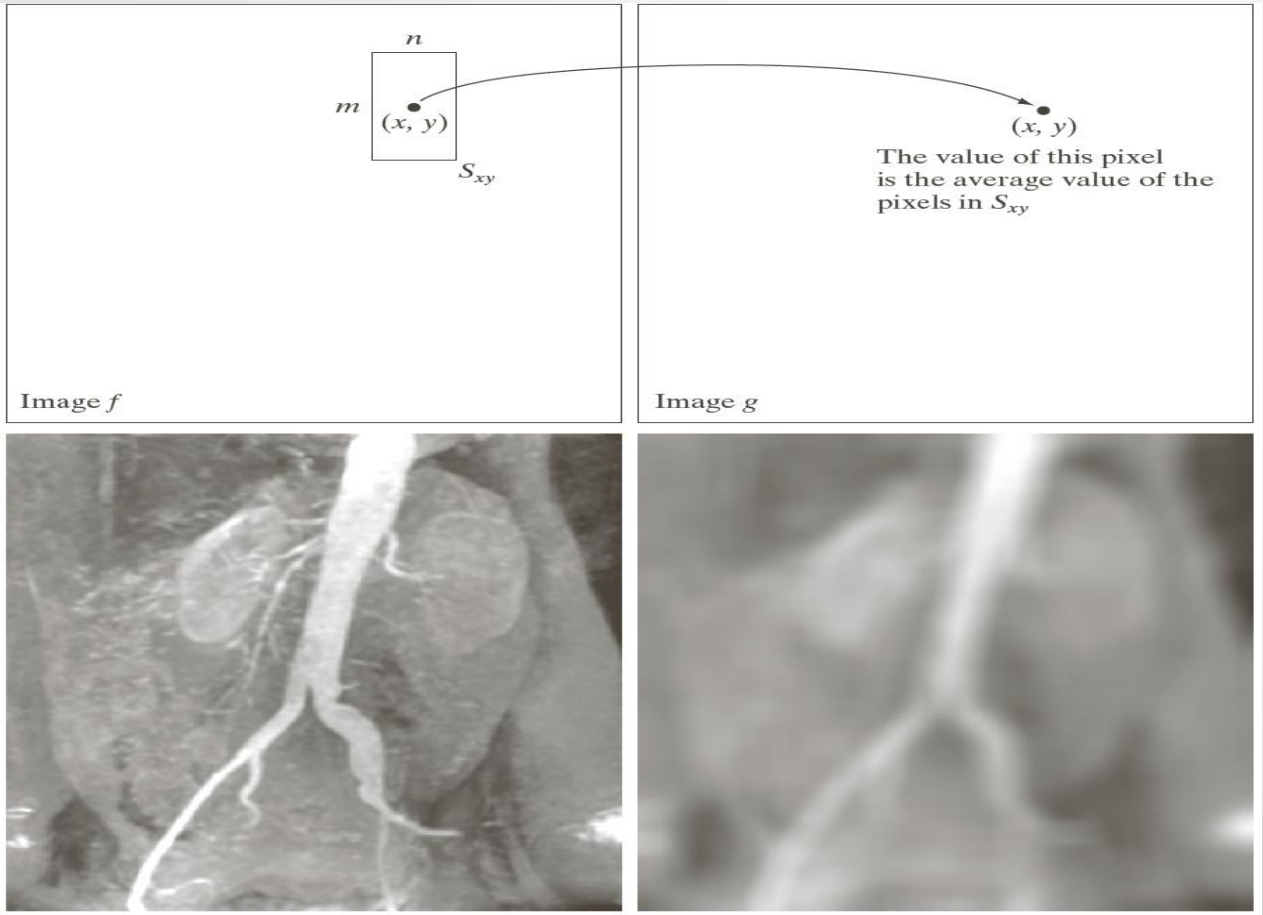
Spatial Domain Image Enhancement

- In addition, T can also work on a set of input images.
- The principal approach to defining a neighborhood about (x, y) is to use a square or rectangular subimage area centered at (x, y), as illustrated below



A 3 x 3 neighborhood about a point (x, y) in an image.

Spatial Domain Image Enhancement (cont')



Local averaging using neighborhood processing.

Spatial Domain Image Enhancement (cont')

- The center of the subimage is moved from pixel to pixel starting, say, at the top left corner and applying the operator at each location (x, y) to yield g at that location.
- Although other neighborhood shapes, such as approximations to a circle, sometimes are used, square and rectangular arrays are by far the most predominant because of their ease of implementation.
- The simplest form of T is when the neighborhood is 1×1. In this case, g *(output)* depends only on the value of f *(input)* at (x, y), and T becomes a *gray-level transformation* (also called *mapping*) *function* of the form

$$s = T(r)$$

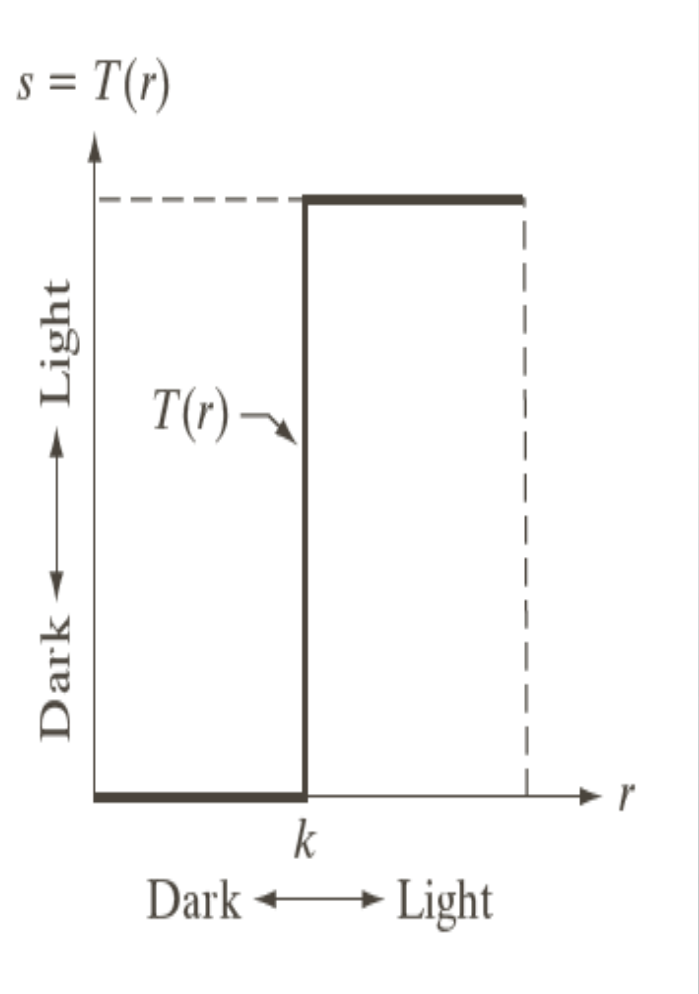
- where, for simplicity in notation, r and s are variables denoting the gray level of $f(x, y)$ and $g(x, y)$ at any point (x, y).

Spatial Domain Image Enhancement (cont')

- In the limiting case shown in figure below, $T(r)$ produces a two-level (binary) image, a.k.a *thresholding* function.

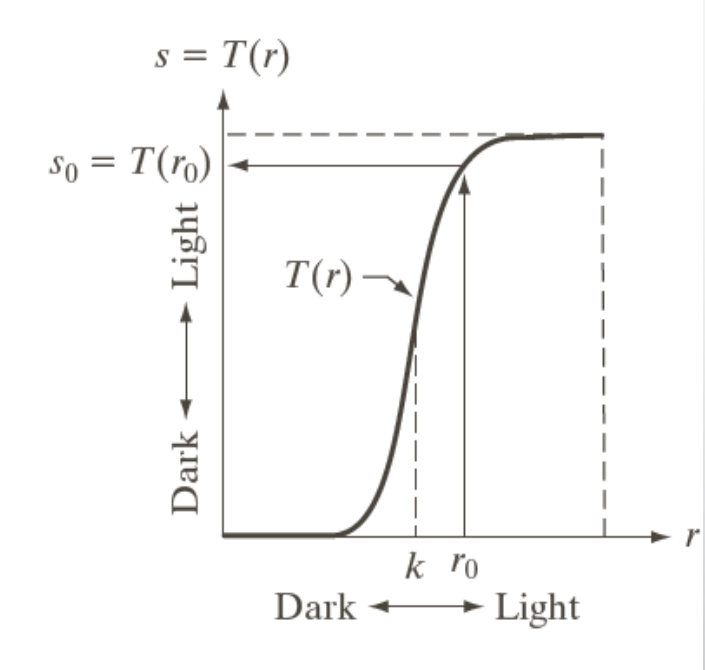
- *Thresholding* converts a grayscale or color image into a binary (black and white) image
- by separating pixels into foreground and background based on a threshold value.
- Pixels with intensity values above the threshold are set to one value (e.g., white), and those below are set to another (e.g., black).
- Because enhancement at any point in an image depends only on the gray level at that point, techniques in this category often are referred to as *point processing (point level)*.
- Point operation includes • *Brightness*

• *Contrast* • *Histogram*



Spatial Domain Image Enhancement (cont')

- For example, if $T(r)$ has the form shown in figure below, the effect of this transformation is to produce an image of higher contrast than the original by darkening the levels below ***k*** and brightening the levels above ***k*** in the original image.



- In this technique, known as *contrast stretching*, the values of r below ***k*** are compressed by the transformation function into a narrow range of s toward black; the opposite effect takes place for values of r above ***k***.
- It is a simple, linear operation that maps the lowest intensity value to the minimum value of the desired range and the highest value to the maximum
- Making details more visible by increasing the differences between dark and bright areas

Spatial Domain **Image Enhancement (cont')**

- **Intensity Histogram**

- The *histogram* of a digital image with gray levels in the range [0, L-1] is a discrete function $h(r_k) = n_k$, where r_k is the kth gray level, n_k is the number of pixels in the image with that gray level.

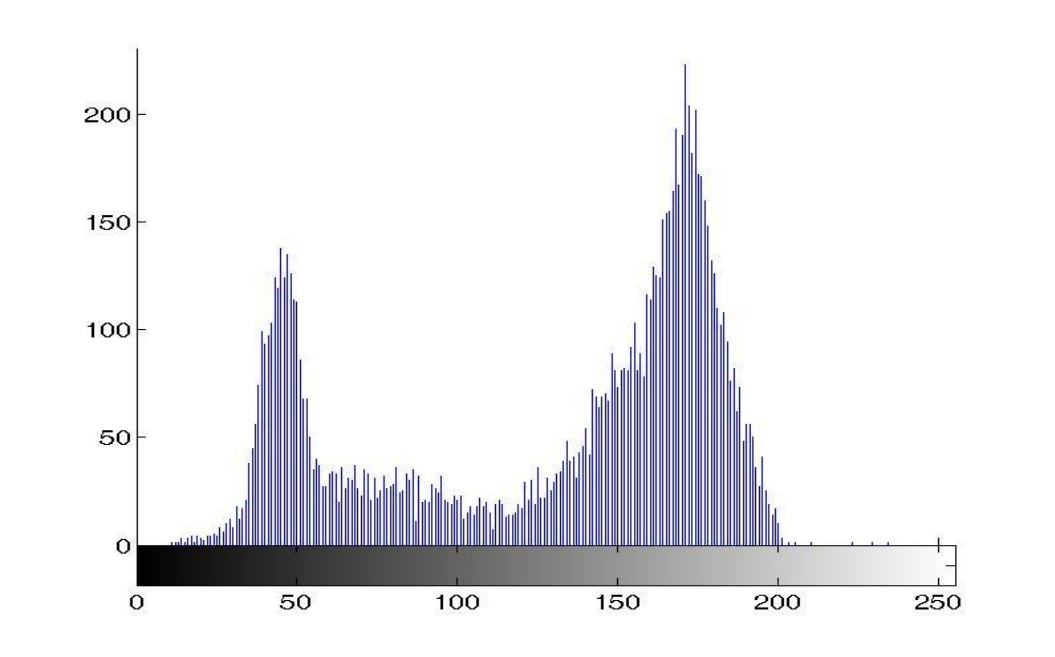
- *Normalized histogram* is given by $p(r_k) = n_k/n$, where n is the total number of pixels in the image, and $k = 0, 1, 2, \ldots, L\text{-}1$.

- $p(r_k)$ is an estimate of the probability of occurrence of intensity level r_k in an image. (Sum of the all component = 1)

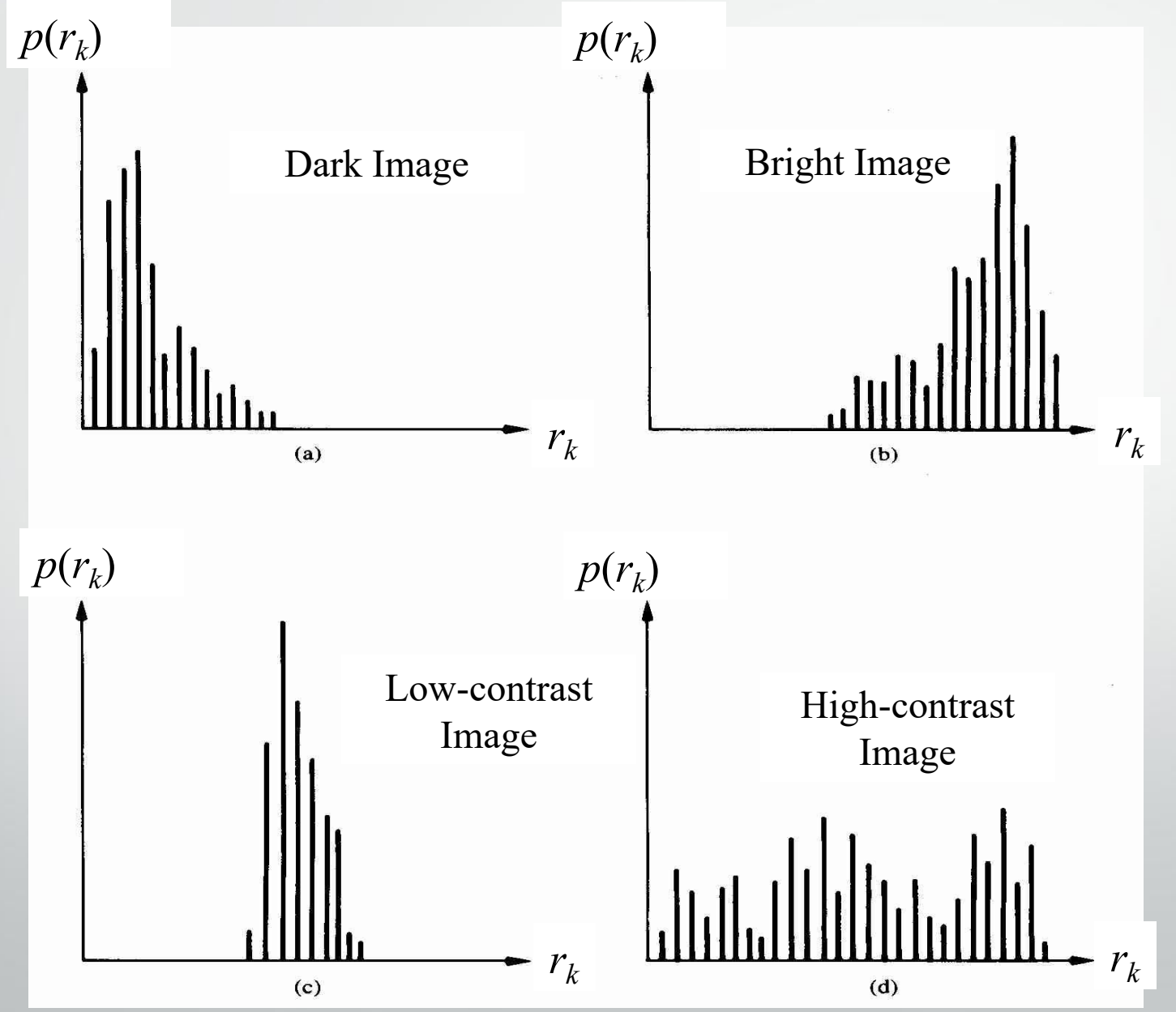
Image Enhancement - Histogram (cont')

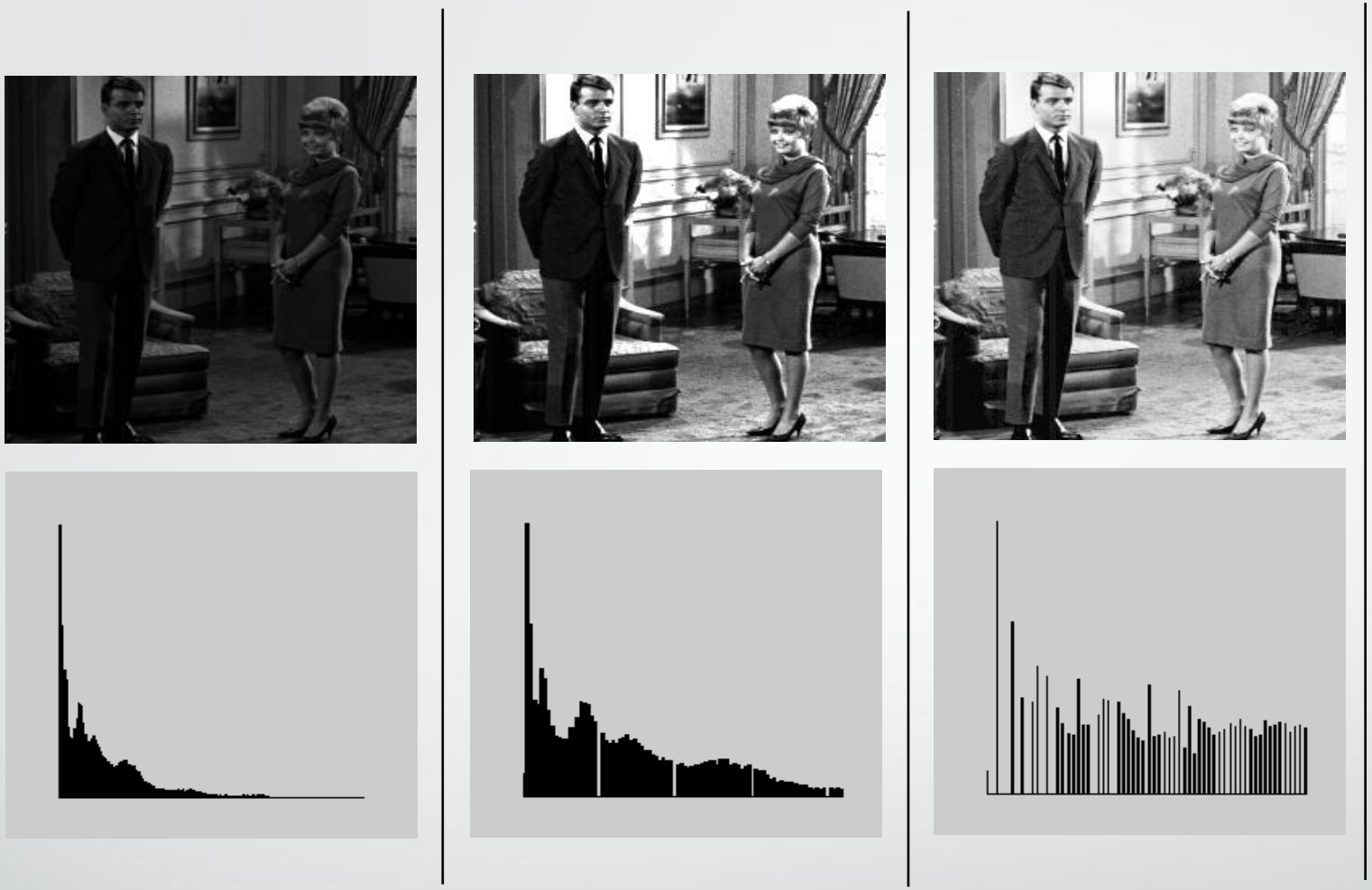
- Histograms are the basis for numerous spatial domain processing techniques. Besides providing useful image statistics, information inherent in histograms is also useful in image processing applications, such as enhancement, compression, and segmentation.
- Histogram of an image, like other histograms, also shows frequency. But an image histogram shows the frequency of pixel intensity values. In an image histogram, the x axis shows the gray level intensities and the y axis shows the frequency of these intensities.

Gray scale image

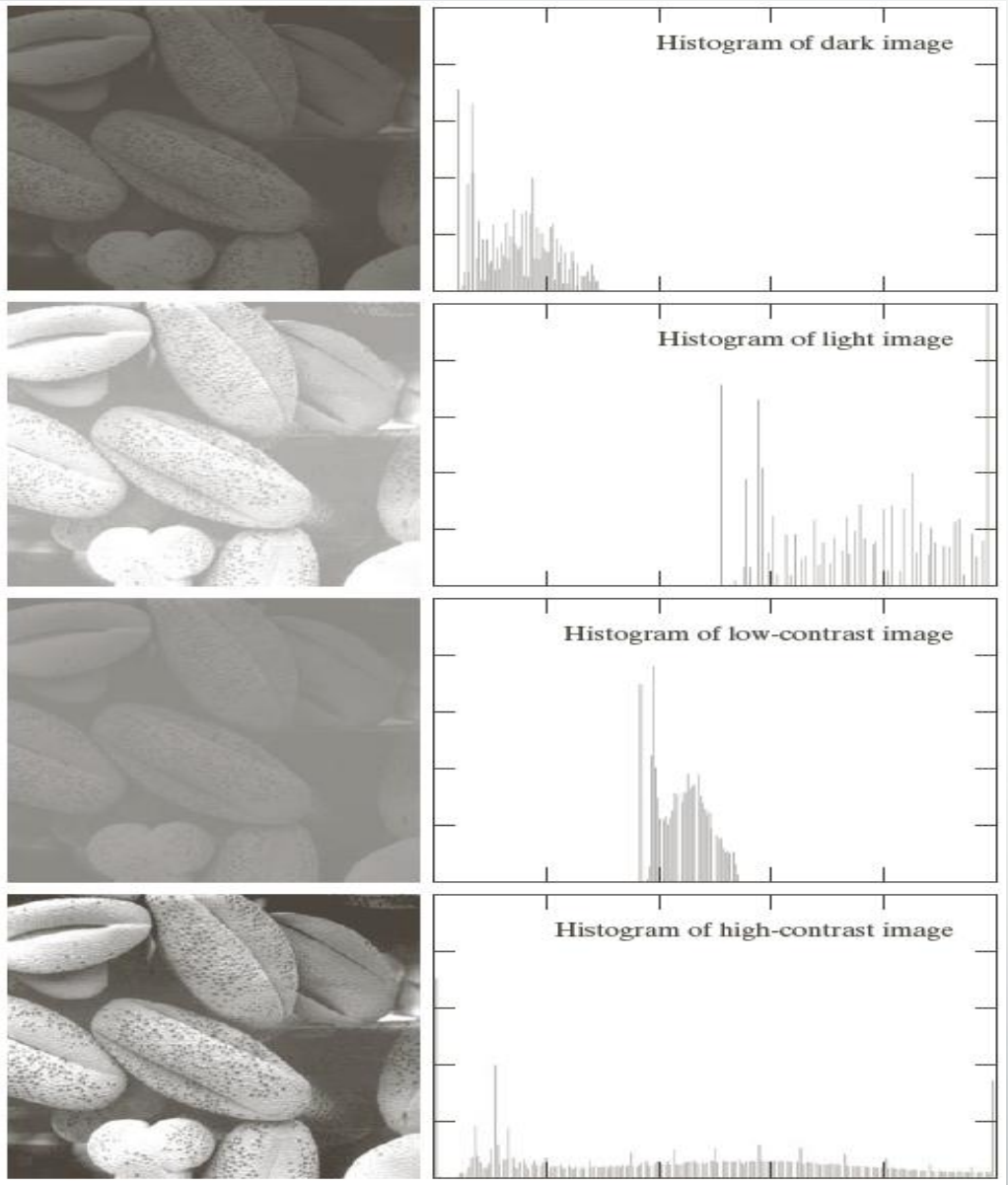


Its intensity histogram


$p(r_k)$
Dark Image
r_k
(a)
$p(r_k)$
Bright Image
r_k
(b)
$p(r_k)$
Low-contrast Image
r_k
(c)
$p(r_k)$
High-contrast Image
r_k
(d)



Although the properties just discussed are global descriptions that say nothing specific about image content, the shape of the histogram of an image does give us useful information about the possibility for contrast enhancement.



Four basic image types: dark, light, low contrast, high contrast, and their corresponding histograms.

Image Enhancement – Thresholding (cont')

- Thresholding is a labeling operation on a gray scale image that distinguishes pixels of a higher intensity from pixels with a lower intensity value
- This technique is particularly useful for scenes which contain solid objects on a uniform contrasting background.
- The output of thresholding is usually a binary image.
- Pixels within the region of interest are given the binary value of 1,and pixels outside of the region of interest are given a binary value of 0.

Image Enhancement Thresholding (cont')

- Thresholding works well when a gray level histogram of the image groups separates the pixels of the object and the background into two dominant modes (bimodal).
- Then a threshold T can be easily chosen between the modes.
- For a light object on a dark background, any pixel for which $f(x,y) > T$, is labeled as an object point. Otherwise the pixel is labeled as a background point.
- Likewise, for a dark object on a light background, any pixel for which $f(x,y) < T$, is labeled as an object point. Otherwise the pixel is labeled as a background point.

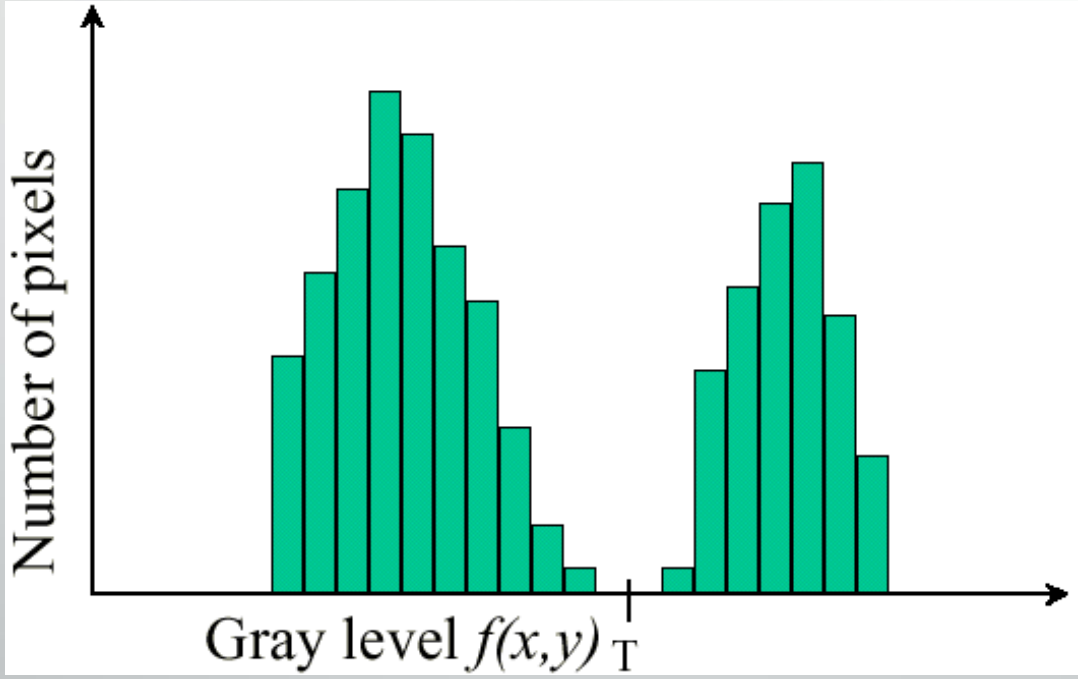
Image Enhancement - Thresholding (cont')

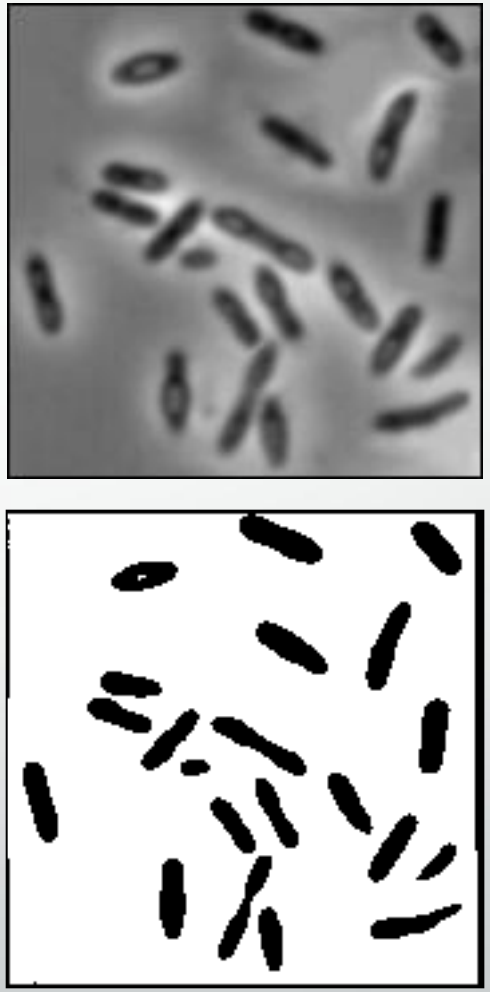
- *Global thresholding* → when T is a constant applicable over an entire image
- Local thresholding → T changes over an image where the value T at any point (x, y) in an image depends on properties of a neighborhood of (x, y) (for example, the average intensity of the pixels in the neighborhood)
- *Dynamic/ adaptive thresholding* → T depends on spatial coordinates x and y
- Thresholded image is defined as:

$$g(x,y) = \begin{cases} 1 & if \;\; f(x,y) > T \\ 0 & if \;\; f(x,y) < T \end{cases}$$

Image Enhancement - *Global thresholding* (cont')

- *Global thresholding*:
  - Choose threshold T that separate object from background





T=102

Image Enhancement (cont')

- The same approach can be used with more than one threshold value.
- In general this technique is less reliable than a single variable threshold. This technique often difficult to establish multiple thresholds to effectively isolate the region of interest.

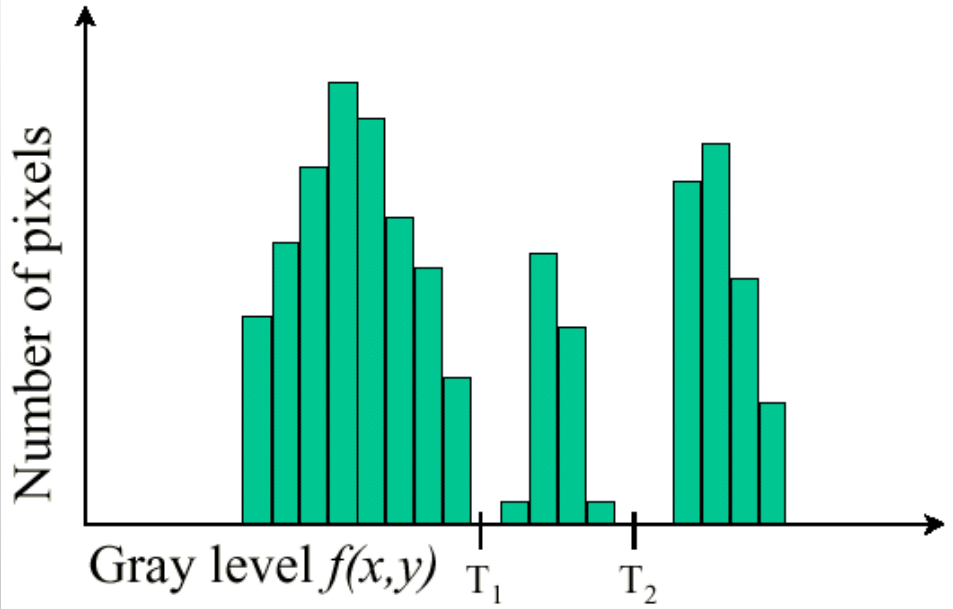


Image Enhancement (cont')

- Simple thresholding is not always possible:
  - Many object at different gray levels
  - Variations in background gray level ⟶ nonuniform illumination
  - Noise in image

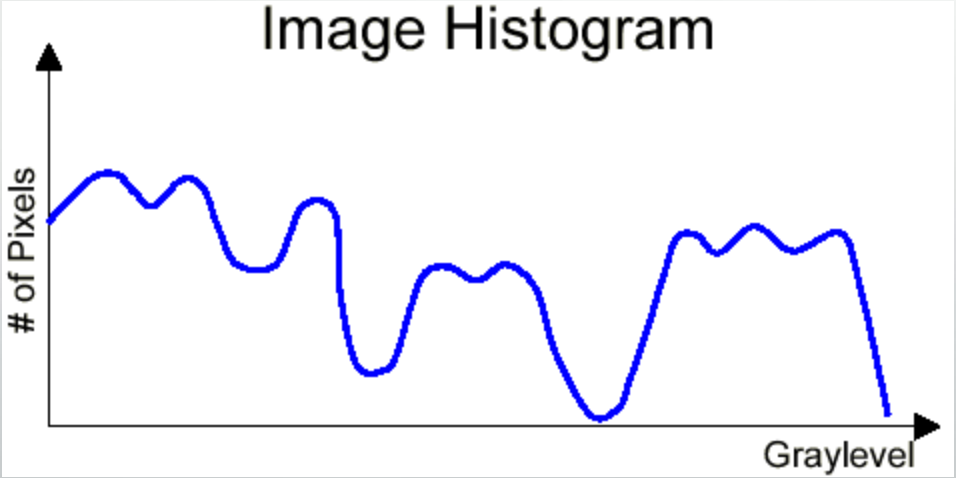
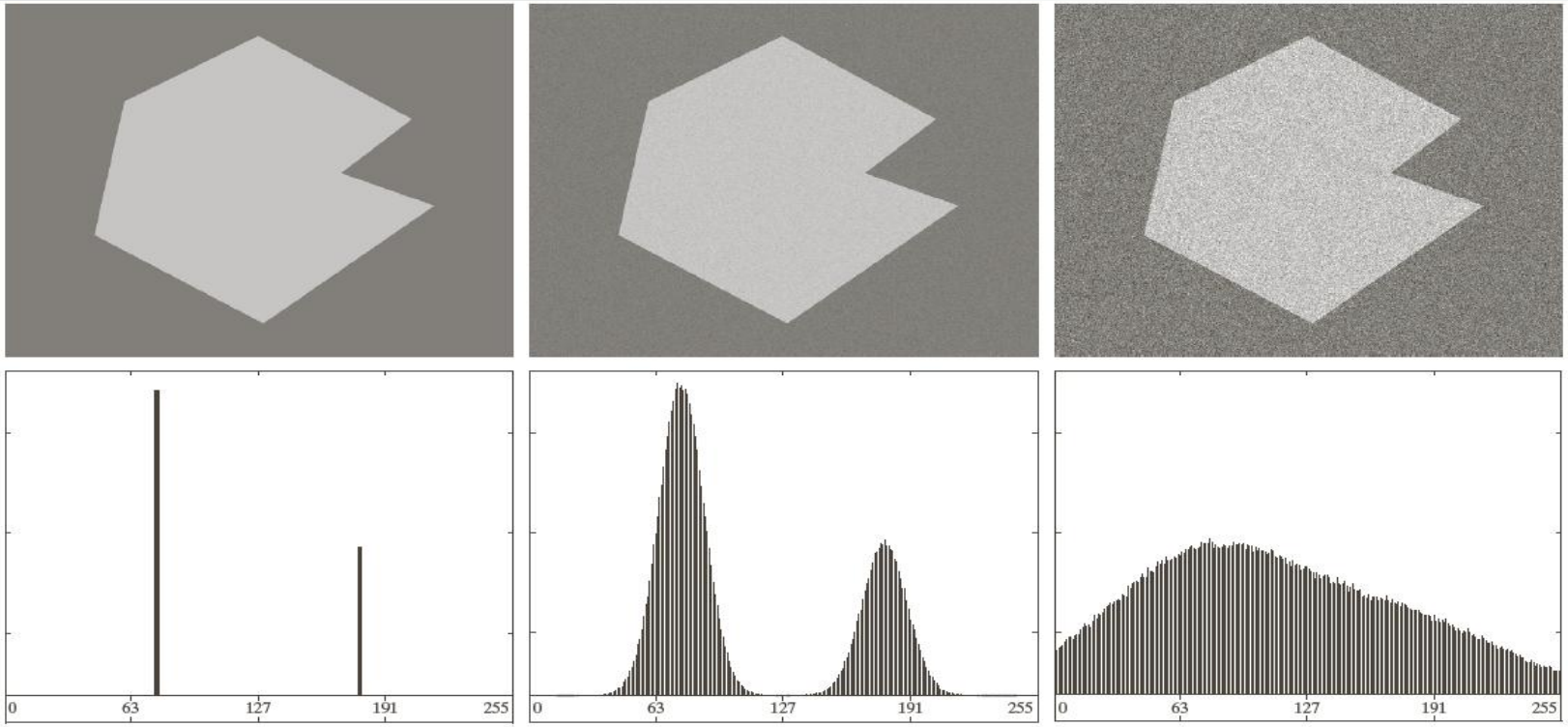


Image Enhancement (cont')



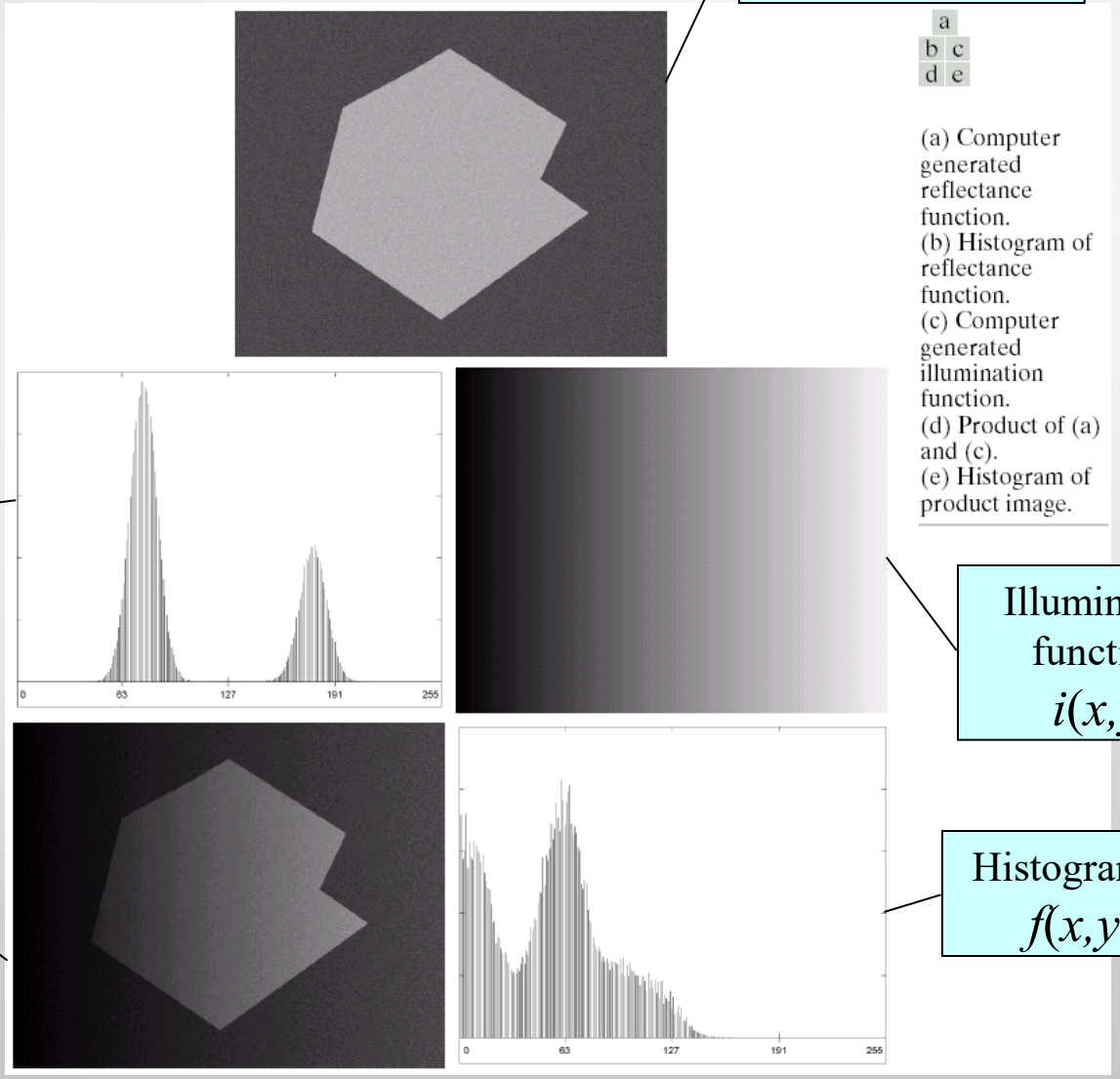
a b c
d e f

(a) Noiseless 8-bit image. (b) Image with additive Gaussian noise of mean 0 and standard deviation of 10 intensity levels. (c) Image with additive Gaussian noise of mean 0 and standard deviation of 50 intensity levels. (d)–(f) Corresponding histograms.

Image Enhancement (cont')

- *Example:*
  - *Nonuniform illumination*

Reflectance function, $r(x,y)$



a
b c
d e

(a) Computer generated reflectance function.
(b) Histogram of reflectance function.
(c) Computer generated illumination function.
(d) Product of (a) and (c).
(e) Histogram of product image.

Histogram of reflectance function

Illumination function, $i(x,y)$

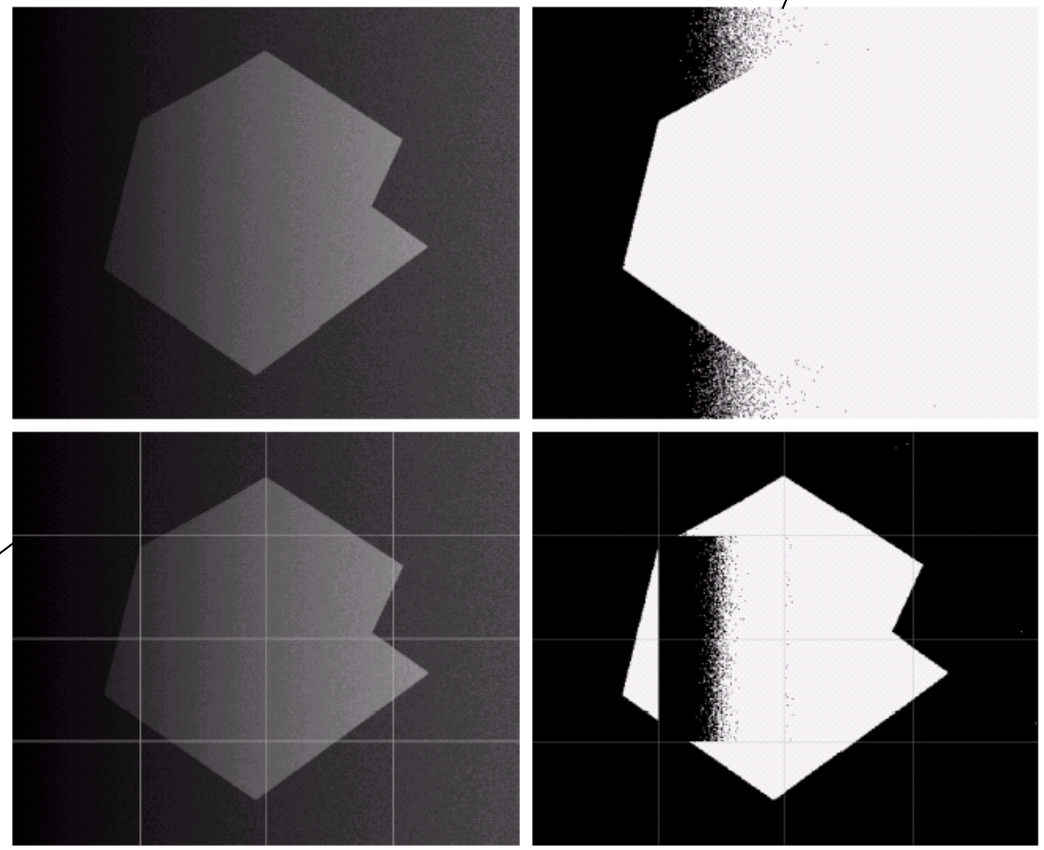
$f(x,y)=i(x,y)r(x,y)$

Histogram of $f(x,y)$

Thresholding (cont')

- *Example: Adaptive thresholding*

Result of global thresholding



a b
c d

FIGURE 10.30 (a) Original image. (b) Result of global thresholding. (c) Image subdivided into individual subimages. (d) Result of adaptive thresholding.

Image divided into subimage such that the illumination of subimage is uniform

Improper segmented regions → no clear distinction between object and background

Image Enhancement **Contrast Stretching** (cont')

- Low-contrast images can result from poor illumination, lack of dynamic range in the imaging sensor, or even wrong setting of a lens aperture during image acquisition.
- The idea behind contrast stretching is to increase the dynamic range of the gray levels in the image being processed.
- Figure (a) on the next page shows a typical transformation used for contrast stretching.
- The locations of points (r_1, s_1) and (r_2, s_2) control the shape of the transformation function.
- For instance, if $r_1 = s_1$ and $r_2 = s_2$, the transformation is a linear function that produces no changes in gray levels. If $r_1 = r_2$, $s_1 = 0$, and $s_2 = L - 1$, the transformation becomes a *thresholding function* that creates a binary image.

Image Enhancement - **Contrast Stretching** (cont')

Contrast stretching: (a) form of transformation function; (b) a low-contrast image; (c) result of contrast stretching; (d) result of thresholding.

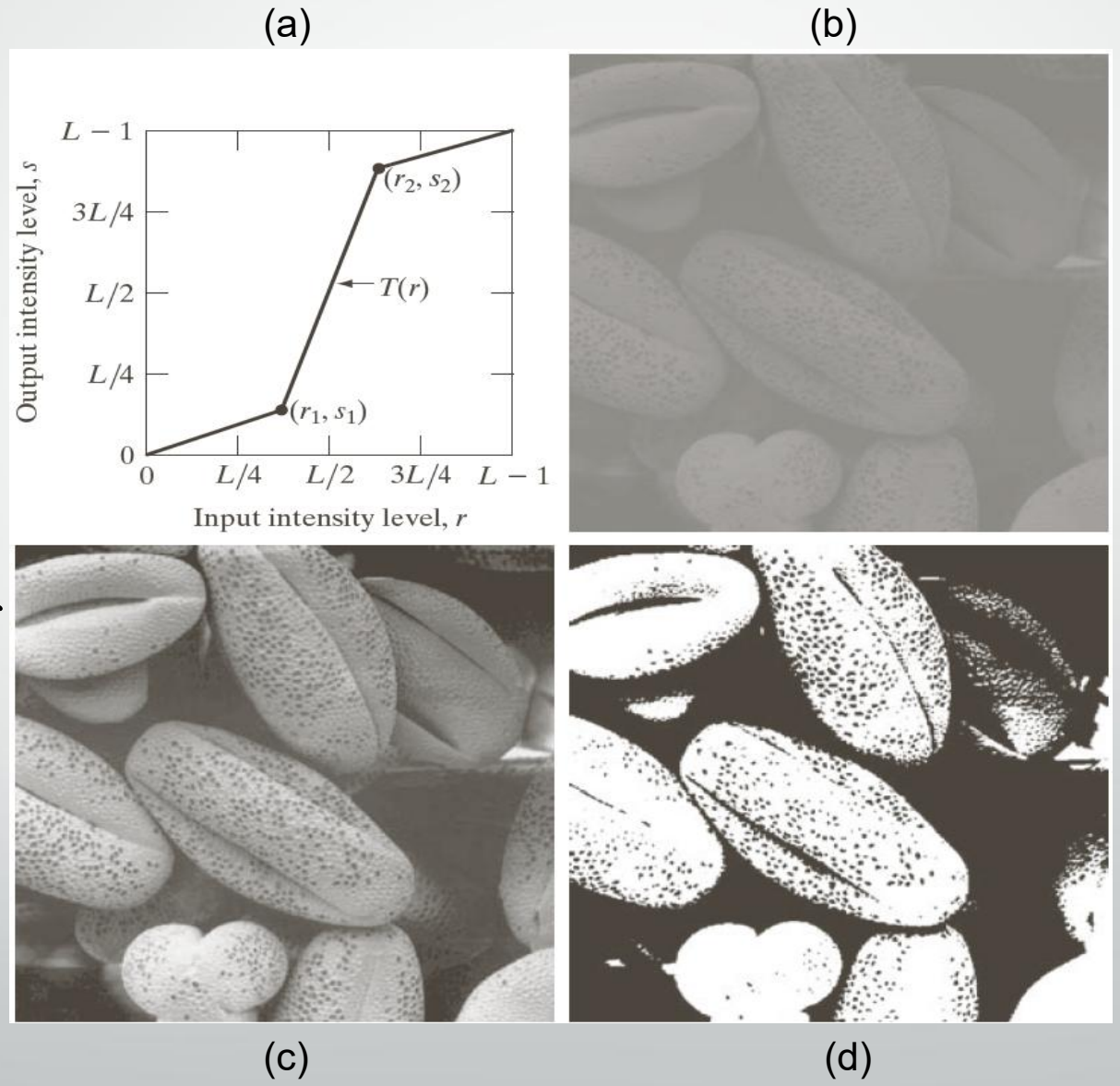


Image Enhancement (cont')

- Intermediate values of (r_1, s_1) and (r_2, s_2) produce various degrees of spread in the gray levels of the output image, thus affecting its contrast.
- In general, $r_1 \leq s_1$ and $r_2 \leq s_2$ assumed so that the function is single valued and monotonically increasing.
- This condition preserves the order of gray levels, thus preventing the creation of intensity artifacts in the processed image.

Image Enhancement- **Image Negatives** (cont')

- Image negatives can be produced with the transformation shown below.

Obtaining the negative of an image: (a) gray-level transformation function; (b) an image; and (c) its negative, in (a), *r* and *s* denote the input and output gray levels, respectively. *L* denotes the total number of gray levels.

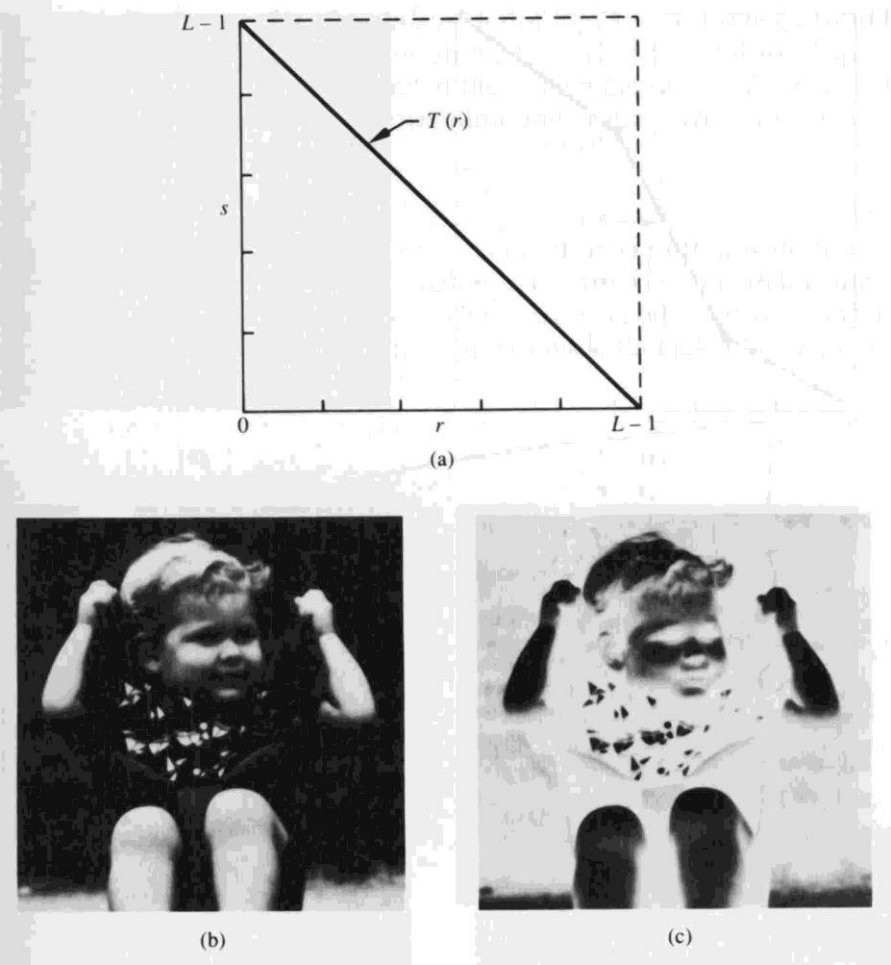


Image Enhancement- **Intensity-level slicing** (cont')

- Highlighting a specific range of gray levels in an image often is desired.
- Applications include enhancing features such as masses of water in satellite imagery and enhancing flaws in *X*-ray images.
- There are several ways of doing level slicing.
- One approach is to display a high value for all gray levels in the range of interest and a low value for all other gray levels.
- This transformation, shown in Figure (a) on the next page, produces a binary image.
- The second approach, based on the transformation shown in Figure (b), brightens the desired range of gray levels but preserves the background and gray-level tonalities in the image.
- Figure (c) shows a gray-scale image, and Figure (d) shows the result of using the transformation in Figure (a).

Image Enhancement (cont')

Refer previous page for description of figure.

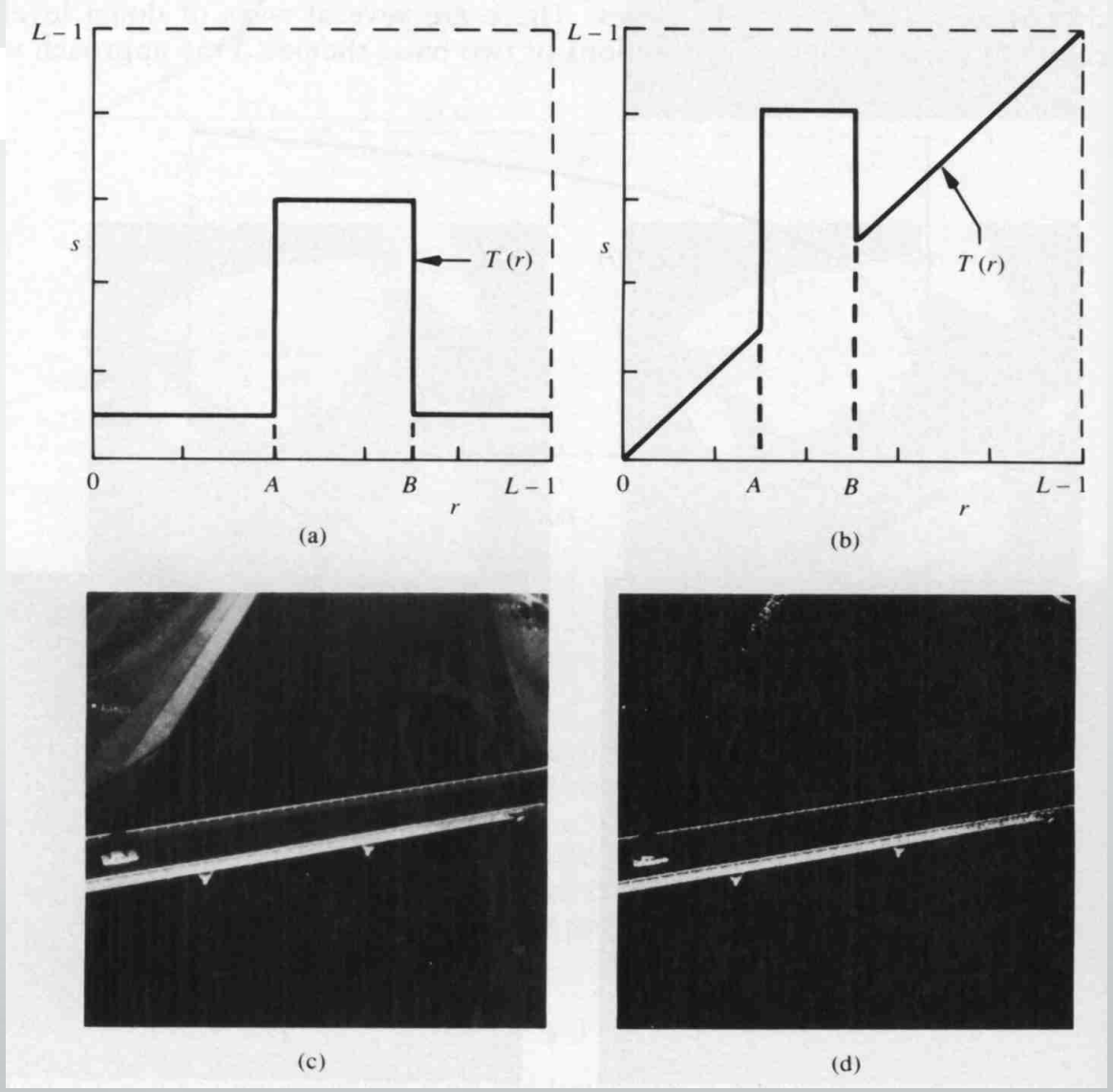
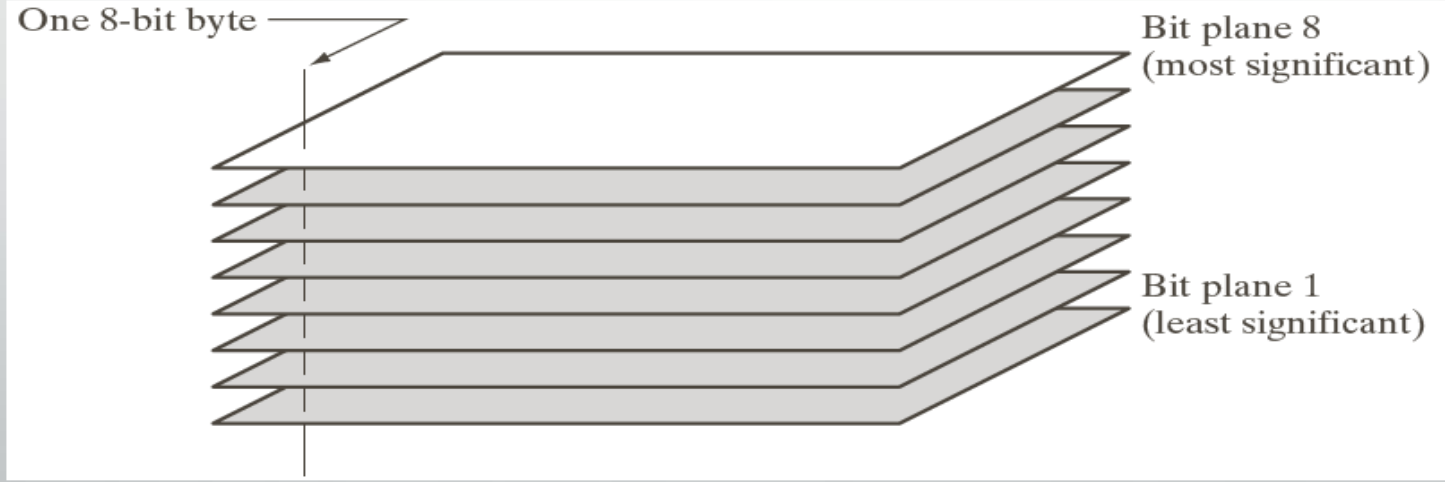
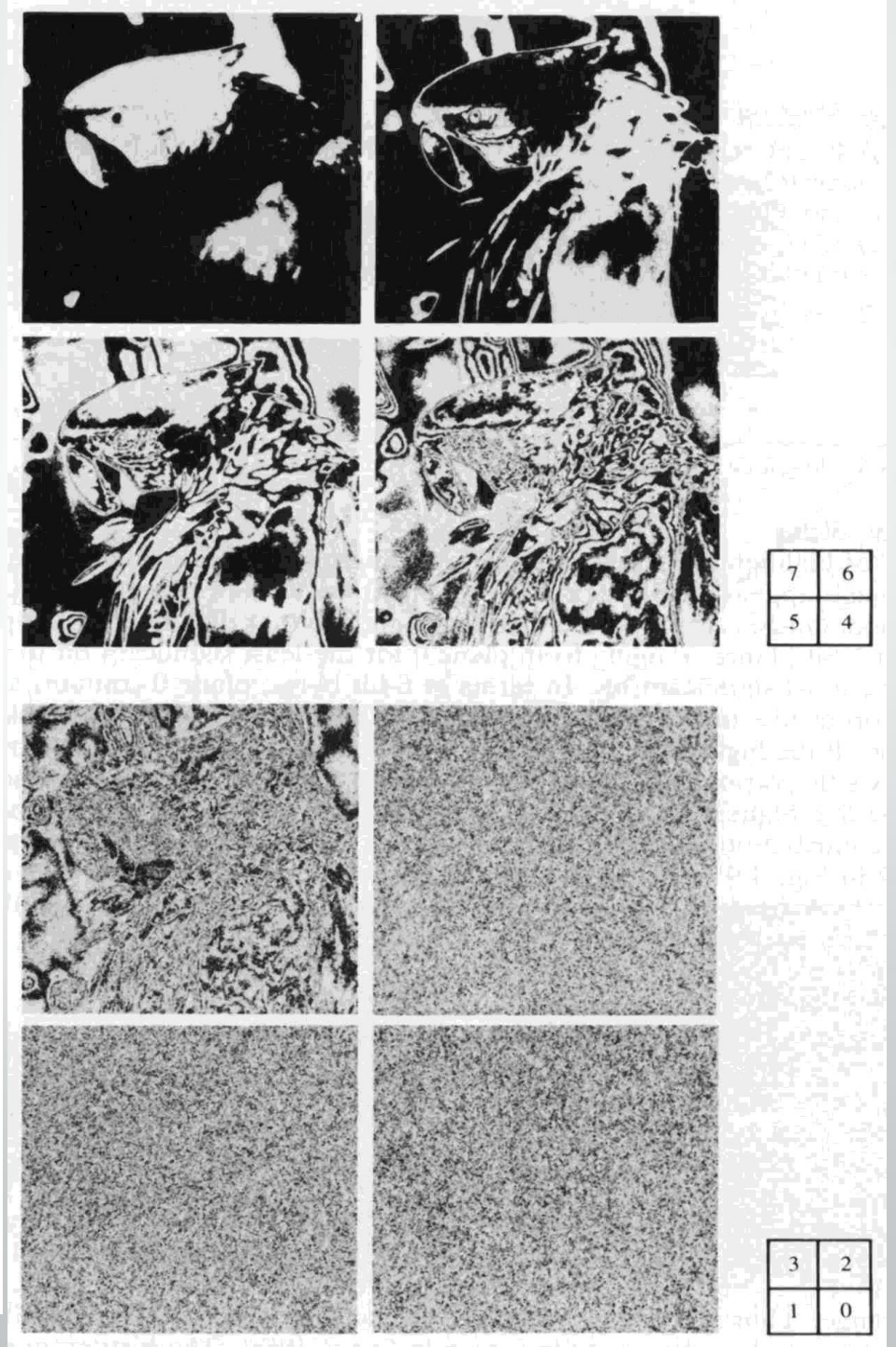


Image Enhancement - **Bit-plane slicing** (cont')

- Instead of highlighting intensity ranges, highlighting the contribution made to total image appearance by specific bits might be desired.
- Suppose that each pixel in an image is represented by 8 bits.
- Imagine that the image is composed of eight 1-bit planes, ranging from plane 0 for the least significant bit to plane 7 for the most significant bit.
- In terms of 8-bit bytes, plane 0 contains all the lowest order bits in the bytes comprising the pixels in the image and plane 7 contains all the high-order bits.



Bit-plane representation for an 8-bit digital image

- Bit planes shown on the left for the image above. The numbers in the small squares identify individual planes.
- Plane 7 contains the most significant bits, and plane 0 contains the least significant bits of the pixels in the original image.